Deep Learning-Based Multiclass Classification of EEG Motor Intentions Using Convolutional Neural Networks

Angel Barrera
Master's Student in Applied Data Science
East Tennessee State University

May 2025

1 Introduction

The interpretation of electroencephalographic (EEG) signals for brain-computer interface (BCI) applications has emerged as a crucial research domain in computational neuroscience and signal processing. Among the most compelling applications of BCI is the decoding of motor intentions from non-invasive EEG data, which enables communication and control for individuals with motor impairments. By detecting user intent through brain activity patterns, BCI systems promise a non-muscular communication pathway with external devices, including prosthetics, robotic arms, and computers.

However, decoding motor intentions from raw EEG signals is inherently challenging due to the signal's high dimensionality, low signal-to-noise ratio, and strong inter-subject variability. These difficulties have motivated the development of robust preprocessing techniques, discriminative feature extraction methods, and powerful classification models capable of learning relevant spatiotemporal patterns from EEG data.

In this project, I focus on EEG recordings collected during motor imagery tasks involving three distinct classes: *rest*, *left fist movement*, and *right fist movement*. Our primary objective is to develop a multiclass classification model that accurately predicts the motor intention underlying each EEG trial. To this end, I implement and compare traditional machine learning approaches, such as Common Spatial Patterns (CSP) combined with Logistic Regression and Principal Component Analysis (PCA), with modern deep learning techniques, particularly a Convolutional Neural Network (CNN) with stacked 1D convolutional layers.

This work uses the publicly available BCI dataset from the BCI Competition IV dataset 2a, hosted on PhysioNet, which provides raw EEG recordings for various motor imagery tasks. Through iterative experimentation and evaluation, our study seeks to identify the most effective strategies for preprocessing, feature engineering, and classification, ultimately contributing insights into the design of practical EEG-based motor intention recognition systems.

2 Problem Statement

The central problem addressed in this study is the reliable classification of motor intentions, specifically, rest, imagined left fist movement, and imagined right fist movement, based on non-invasive EEG signals. While EEG provides a cost-effective and temporally precise method for recording brain activity, its inherent limitations, including low spatial resolution, signal variability across sessions and subjects, and susceptibility to noise, make accurate decoding of motor imagery a non-trivial task.

Traditional approaches such as Common Spatial Patterns (CSP) and linear classifiers have shown promise in binary tasks but often struggle to generalize in multiclass settings and across individuals. This project aims to overcome those limitations by leveraging deep learning architectures that can directly learn discriminative features from raw or minimally preprocessed EEG data.

The challenge lies in developing a robust model that not only achieves high classification accuracy on held-out validation data but also demonstrates interpretability, resilience to class imbalance, and potential for real-time inference, key requirements for real-world BCI applications.

3 Motivation

The ability to decode motor intentions from EEG signals has transformative implications for individuals with severe motor disabilities. By enabling hands-free, intention-driven control of external devices, brain-computer interfaces (BCIs) offer a pathway to restore autonomy and improve quality of life. However, realizing this potential requires overcoming several persistent challenges in EEG signal analysis, including low signal-to-noise ratio, inter-subject variability, and the non-stationary nature of brain activity.

This project is motivated by the opportunity to push beyond traditional feature-engineering pipelines by applying deep learning models that can learn discriminative spatiotemporal features directly from raw EEG data. Specifically, I aim to improve upon existing binary motor imagery classification approaches by exploring a data-driven framework using 1D convolutional neural networks.

In doing so, I not only contribute toward improving classification accuracy in controlled experimental settings but also help lay the groundwork for real-time, non-invasive BCI systems capable of translating brain activity into actionable commands, particularly for prosthetic control or assistive communication systems.

4 Research Questions

This project aims to answer the following research questions:

1. Can deep learning models, specifically 1D Convolutional Neural Networks, effectively classify motor imagery EEG signals corresponding to imagined left and right fist movements?

2. How does the performance of deep learning models compare to traditional methods such as Common Spatial Patterns (CSP) combined with logistic regression or PCA-based classifiers?

3. Does excluding the dominant 'rest' (T0) class from the training set improve classification performance and generalization in motor intention decoding?

4. Which EEG frequency bands and spatial regions contribute most to distinguishing between left and right motor imagery?

5 Literature Review

Research on decoding motor imagery from EEG signals has long focused on transforming raw, noisy data into informative features using domain-specific techniques. One of the most widely used methods is *Common Spatial Patterns* (CSP), which projects multichannel EEG data into a subspace that maximizes variance differences between two motor imagery classes. CSP combined with linear classifiers, such as logistic regression or support vector machines (SVMs), has shown moderate success in binary tasks but often fails to generalize in complex or noisy scenarios [Ramoser et al., 2000].

To overcome the limitations of handcrafted features, deep learning models, particularly Convolutional Neural Networks (CNNs), have gained traction. Lawhern et al. introduced *EEGNet*, a compact CNN architecture designed for EEG-based BCIs, demonstrating that CNNs can extract meaningful spatiotemporal features directly from raw signals [Lawhern et al., 2018]. Later efforts incorporated recurrent structures like LSTMs to model temporal dependencies, further improving performance in sequence modeling tasks.

A relevant example is an open-source project by Reshal Fahsi [Fahsi, 2020], which compares CNN and RNN (LSTM) architectures for classifying EEG motor imagery data using the BCI Competition IV 2a dataset. Their approach, while not tailored for real-time BCI deployment, confirms the effectiveness of deep learning pipelines over traditional classifiers and highlights the importance of model architecture design and preprocessing choices.

Additionally, studies have emphasized the role of frequency bands, especially Mu (8–12 Hz) and Beta (13–30 Hz),in motor-related EEG activity [Pfurtscheller and Lopes da Silva, 1999], as well as the potential of channel selection and subject-specific adaptation to improve generalization across users.

Building on this foundation, our work implements and compares traditional CSP pipelines with a deep CNN architecture trained on filtered and normalized EEG data. Our goal is to evaluate which method yields the most robust results in binary classification of imagined left vs. right hand movements.

6 Dataset Description

This project uses EEG data from the publicly available *PhysioNet BCI Motor Movement/Imagery Dataset* [Goldberger et al., 2000], originally recorded using the BCI2000 system [Schalk et al., 2004]. The dataset comprises electroencephalogram (EEG) recordings from 109 subjects who performed a series of real and imagined motor tasks. The dataset was sourced from the PhysioNet platform

Structure and Acquisition

Each subject completed 14 experimental runs involving:

- **Resting-state sessions** (eyes open or closed)
- **Motor execution tasks** (real movement of fists or feet)
- **Motor imagery tasks** (imagined movement of left fist, right fist, both fists, or both feet)

Recordings were obtained using a 64-channel EEG cap with a sampling rate of 160 Hz. Data is stored in EDF format, with annotations marking the beginning of each task trial using specific event codes:

- **T0**: Rest
- **T1**: Imagined movement of the left fist (or both fists, depending on run)
- **T2**: Imagined movement of the right fist (or both feet)

Figures A1, A2, and A3 visualize example EEG trials for each of these classes. These plots help illustrate the differences in amplitude and structure between rest and motor imagery signals.

Selection for This Study

For this study, data was extracted from 20 subjects (S001–S020), focusing on runs 4, 8, and 12, which correspond to motor imagery trials. These runs contain the imagined movements relevant to our classification objective.

Final Dataset

After preprocessing and label filtering:

- A total of 900 trials were included across all subjects.
- Class distribution: 426 trials labeled as `T1` (left fist) and 423 trials as `T2` (right fist).

- The `T0` (rest) class was excluded from the final model to reduce class imbalance and focus exclusively on motor intention decoding between active classes.

The rationale behind excluding the rest class will be further discussed in the Results section, where I evaluate the impact of this decision on model performance and generalization. Having the resulting dataset provided a balanced and clean sample for supervised learning, focused specifically on decoding motor intentions between two opposing imagery classes.

7 Approach

Our approach to EEG motor intention classification followed a structured pipeline, progressing through data extraction, preprocessing, feature design, and model development. The goal was to build a reliable binary classifier for imagined left and right fist movements.

7.1 Data Collection

The dataset was sourced from the PhysioNet BCI Motor Movement/Imagery database, containing EEG recordings in EDF format. I selected motor imagery sessions from 20 subjects (S001–S020), specifically runs 4, 8, and 12, which include annotated segments for left and right fist movements. A total of 900 labeled trials were extracted from these recordings.

7.2 Data Cleaning

I performed the following preprocessing steps:

- **Bandpass Filtering:** Signals were filtered between 1–40 Hz to remove low-frequency drift and high-frequency noise.
- **Trial Segmentation:** Trials were extracted between 1s and 4.1s post-cue, yielding 497 time samples per trial.
- **Z-Score Normalization:** Each EEG channel was standardized across time to have zero mean and unit variance.
- **Label Filtering:** Only T1 (left) and T2 (right) labels were retained to ensure class balance and focus the model on motor intention decoding.

7.3 Feature Extraction

I explored two approaches to feature extraction:

- **Manual Features:** Using Common Spatial Patterns (CSP) to transform the EEG data into spatially discriminative features.
- **Learned Features:** Using 1D Convolutional Neural Networks (CNNs) to automatically extract temporal and spatial features from the raw signal.

7.4 Feature Selection

For the CSP-based methods, I selected the top components (4–16) that maximized variance differences between classes. In deep learning models, no explicit feature selection was performed; instead, the network was trained end-to-end to learn relevant filters.

7.5 Model Building

Evaluating both traditional and deep learning classifiers:

- **Traditional Models:** CSP + Logistic Regression, CSP + PCA + Random Forest/SVM. These models achieved accuracies between 47% and 54%.
- **Deep Learning Models:** Our final model was a CNN with two stacked Conv1D layers, ReLU activation, batch normalization, and a softmax output layer. It achieved a validation accuracy of **76%**, outperforming all prior methods.

The model was trained using categorical cross-entropy loss and evaluated on a held-out validation set using accuracy, ROC-AUC, and precision-recall metrics.

7.6 Results and Insights

A range of models for classifying imagined left and right fist movements (T1 vs. T2) were evaluated based on EEG data from 20 subjects. Key findings are summarized below by model type.

7.6.1 CSP + Logistic Regression

The Common Spatial Patterns (CSP) algorithm was used to extract features by maximizing the variance difference between two classes in the EEG signal space. I tested CSP with 8, 12, and 16 components, each followed by a Logistic Regression classifier to evaluate performance on binary classification (Left vs. Right).

Figure A4 illustrates the core architecture and result of this pipeline.

Accuracy: The performance remained close to random chance across all CSP component configurations:

- **8 components:** Accuracy = 50.00%
- **12 components:** Accuracy = 44.81%
- **16 components:** Accuracy = 49.53%

Classification Report (8 components):

- **Precision:** Left = 0.50, Right = 0.51
- **Recall:** Left = 0.55, Right = 0.46

- **F1-score:** Left = 0.52, Right = 0.48
- **Accuracy:** 50.0%

Figure A5 and Figure A6 illustrate CSP component behavior and regularization effects.

7.6.2 CSP + PCA + Traditional Classifiers

In this approach, CSP was first used to extract spatial features from the EEG data. These features were then compressed using Principal Component Analysis (PCA) to reduce dimensionality before being passed into traditional classifiers: Random Forest, Gradient Boosting, and Support Vector Machine (RBF kernel).

Classifier 1: Random Forest The Random Forest model trained on PCA-compressed CSP features achieved:

- **Accuracy:** 48%
- **Precision:** Left = 0.48, Right = 0.48
- **Recall:** Left = 0.45, Right = 0.51
- **F1-score:** Left = 0.47, Right = 0.50

The confusion matrix showed relatively even but weak performance:

$$\text{Confusion Matrix: } \begin{bmatrix} 48 & 58 \\ 52 & 54 \end{bmatrix}$$

Figure A7 visualizes these results.

Classifier 2: Gradient Boosting Gradient Boosting performed similarly:

- **Accuracy:** 48%
- **Precision:** Left = 0.48, Right = 0.48
- **Recall:** Left = 0.49, Right = 0.47
- **F1-score:** Left = 0.49, Right = 0.48

$$\text{Confusion Matrix: } \begin{bmatrix} 52 & 54 \\ 56 & 50 \end{bmatrix}$$

Figure A8 provides the corresponding output.

Classifier 3: SVM (RBF Kernel) Support Vector Machines with an RBF kernel underperformed relative to expectations:

- **Accuracy:** 47%
- **Precision:** Left = 0.48, Right = 0.45
- **Recall:** Left = 0.65, Right = 0.28
- **F1-score:** Left = 0.55, Right = 0.35

$$\text{Confusion Matrix: } \begin{bmatrix} 69 & 37 \\ 76 & 30 \end{bmatrix}$$

Figure A9 illustrates the imbalance.

7.6.3 EEGNet-Inspired Fully Connected Network

The first deep learning approach was developing a dense neural network inspired by the EEGNet architecture, aiming to classify motor intentions from flattened EEG signals. This model consisted of several fully connected (dense) layers with ReLU activations and dropout regularization. Unlike EEGNet's spatial-temporal filters, this architecture operated on raw, vectorized EEG input.

Model Architecture

- Input: Flattened 2D EEG trial matrix (channels $\times$ timepoints) into 1D vector
- 2 Dense layers (ReLU activations)
- Dropout layers for regularization
- Softmax output for binary classification

Performance Summary

- **Accuracy:** 29.03%
- **Precision:** Left = 0.24, Right = 0.35
- **Recall:** Left = 0.30, Right = 0.29
- **F1-score:** Left = 0.27, Right = 0.31

$$\text{Confusion Matrix: } \begin{bmatrix} 32 & 74 \\ 60 & 46 \end{bmatrix}$$

This confusion matrix shows a heavy misclassification bias across both classes, indicating that the model could not separate the classes above chance.

7.6.4 Convolutional Neural Network (CNN) + Transformer Hybrid

To capture both spatial and temporal dependencies in EEG data, I implemented a hybrid model combining Convolutional Neural Networks (CNN) with a Transformer encoder block. This architecture was designed to first extract low-level features using Conv1D layers, followed by self-attention mechanisms to model long-range dependencies across time.

Mathematical Representation Let $\mathbf{X} \in R^{C \times T}$ be the input EEG segment with C channels and T time samples. The CNN layer learns temporal filters:

$$\mathbf{H}_1 = \text{ReLU}(\text{Conv1D}(\mathbf{X}))$$

This output is then enriched with positional encodings and passed into a Transformer encoder:

$$\mathbf{H}_2 = \text{TransformerEncoder}(\mathbf{H}_1 + \text{PosEnc})$$

The final prediction is obtained via a softmax output layer after dense projection:

$$\hat{y} = \arg\max_k \ \sigma(\mathbf{W}_o \cdot \mathbf{H}_2 + \mathbf{b}_o)_k$$

where $\sigma(\cdot)$ is the softmax function and $k \in \{0, 1\}$.

Training minimizes the binary categorical cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{2} y_{i,k} \log(\hat{y}_{i,k})$$

Model Architecture

- 1 Conv1D layer with ReLU activation
- Positional encoding added to convolutional output
- Transformer encoder block with multi-head self-attention
- Dense layers and dropout for classification
- Softmax output layer for binary prediction (T1 vs T2)

Figure A10 shows a visualization of this architecture.

Performance Summary

- **Accuracy:** 67.74%
- **Precision:** Left = 0.62, Right = 0.73
- **Recall:** Left = 0.83, Right = 0.49
- **F1-score:** Left = 0.71, Right = 0.59

$$\text{Confusion Matrix: } \begin{bmatrix} 69 & 14 \\ 43 & 41 \end{bmatrix}$$

7.6.5 Convolutional Neural Network (CNN) with Two Conv1D Layers

After iterative experimentation, the final and best-performing model utilized a simple yet effective architecture based on stacked Conv1D layers. This model was designed to directly learn discriminative spatiotemporal features from raw EEG signals without the need for manual feature engineering.

Mathematical Representation Let $\mathbf{X} \in R^{C \times T}$ represent the input EEG trial. The model applies two convolutional layers:

$$\mathbf{H}_1 = \text{ReLU}(\text{BN}(\text{Conv1D}_1(\mathbf{X}))), \quad \mathbf{H}_2 = \text{ReLU}(\text{BN}(\text{Conv1D}_2(\mathbf{H}_1)))$$

Next, global average pooling is applied across the temporal axis:

$$\mathbf{z} = \text{GAP}(\mathbf{H}_2)$$

The final class prediction is:

$$\hat{y} = \arg\max_k \; \sigma(\mathbf{W} \cdot \mathbf{z} + \mathbf{b})_k$$

The model is trained to minimize the same cross-entropy loss as above:

$$\mathcal{L} = -\sum_{i=1}^{N} \sum_{k=1}^{2} y_{i,k} \log(\hat{y}_{i,k})$$

Model Architecture

- Input: 64 EEG channels $\times$ 497 time samples
- 2 Conv1D layers with ReLU activation and batch normalization
- Global average pooling
- Dense layer with softmax activation for binary classification

This architecture balanced depth and regularization, enabling the network to generalize across EEG variability from 20 subjects. Figure A11 presents the architecture design.

Performance Summary

- **Accuracy:** 76.61%
- **Precision:** Left = 0.71, Right = 0.82
- **Recall:** Left = 0.87, Right = 0.64
- **F1-score:** Left = 0.78, Right = 0.72

$$\text{Confusion Matrix:} \begin{bmatrix} 23 & 3 \\ 8 & 14 \end{bmatrix}$$

As seen in Figure A12, the model classified left-hand trials more reliably than right-hand ones, though both achieved strong recall and precision.

Evaluation Curves

- **ROC AUC:** 0.79, indicating solid separability between the classes.
- **Average Precision:** 0.83, reflecting a robust model under imbalanced conditions.

Figures A13 and A14 show the ROC and PR curves, respectively.

8 Answers to the Research Questions

1. **Can EEG signals be used to reliably classify different motor intentions (rest, left, and right movement)?**

   Initially, I considered all three classes: `T0` (rest), `T1` (left fist), and `T2` (right fist). However, due to strong class imbalance and low discriminatory power observed for rest trials (as shown in Figure A15), I excluded T0 and focused on the binary classification of T1 vs. T2. Within this setup, deep learning models demonstrated that EEG signals do indeed carry enough discriminative information to classify motor intentions with high accuracy, with the best model achieving **76.61% accuracy** on the validation set of 45 unseen trials.

2. **Which machine learning models perform best in decoding motor imagery from EEG data?**

   Across all models evaluated, traditional classifiers (CSP + Logistic Regression, PCA + Random Forest/Gradient Boosting/SVM) failed to exceed 50%–54% accuracy, indicating weak generalization. Deep learning models, particularly those using convolutional layers, outperformed all baselines. The final CNN with two Conv1D layers achieved the best results, offering a substantial performance margin over both traditional models and the CNN+Transformer hybrid.

3. **What preprocessing and feature extraction techniques enhance classification performance?**

   Preprocessing steps that improved performance included:

   - Filtering EEG signals to remove noise and standardize channel input.
   - Normalization across trials and subjects.
   - Exclusion of the rest class (T0) to address label imbalance.
   - Using raw EEG windows as input to CNNs, which allowed the network to learn temporal patterns directly instead of relying on handcrafted features.

   Feature extraction using CSP provided only modest improvement when used alone. However, learned feature representations via convolutional filters proved more robust.

4. **Does removing the rest class (T0) from the dataset improve classification performance?**

   Yes. Removing T0 significantly improved both class balance and model focus. When all three classes were included, the rest class dominated the model's predictions due to its larger sample size and lower signal complexity (Figure A15). By focusing on T1 and T2 only, the classifier could better learn discriminative temporal dynamics relevant to intentional movements, boosting accuracy by over 25 percentage points.

5. **Can a deep learning model generalize well across multiple subjects in a motor imagery classification task?**

   While the final CNN generalized well on the pooled validation set, cross-subject generalization showed variability. Subject-wise validation accuracy ranged from 42% to 58%, indicating that individual brain patterns differ considerably. Still, the model achieved an average accuracy of 51.8% across 20 subjects, outperforming traditional classifiers and showing promising potential with further subject adaptation or personalization.

9 Conclusion

This study set out to investigate whether EEG signals could be used to accurately classify motor intentions, specifically, imagined left and right fist movements, using both traditional and deep learning-based approaches. The research questions guiding this work centered on model effectiveness, the role of preprocessing, the impact of class balancing, and the feasibility of subject-general models. The results offer clear evidence that EEG signals contain class-separable information that, when appropriately processed and modeled, can support robust motor intention decoding.

Traditional pipelines such as CSP combined with Logistic Regression, as well as PCA-driven variants with Random Forest, Gradient Boosting, and SVM classifiers, consistently underperformed. Their accuracy fluctuated around chance levels (47–50%), revealing that linear and shallow classifiers fail to capture the nonlinear spatiotemporal dynamics inherent in EEG signals. These findings answer the second research question definitively: conventional techniques provide a foundational benchmark, but are inadequate for this type of neural decoding task.

The deep learning experiments demonstrated clear improvements. The EEGNet-inspired fully connected network failed to converge due to its simplistic architecture and the flattening of the input, which discarded critical temporal and spatial information. However, the hybrid CNN + Transformer model revealed that introducing temporal attention mechanisms enables the model to extract more context-aware features, reaching 67.74% accuracy. Despite this improvement, its uneven class performance, favoring left-hand imagery, highlighted a remaining sensitivity to class imbalance and training dynamics.

Ultimately, the final CNN architecture with two Conv1D layers delivered the strongest results. Achieving a validation accuracy of **76%** on a carefully held-out set of 45 balanced trials, the model demonstrated consistent discriminative power. With supporting metrics including a ROC AUC of 0.79 and an average precision of 0.83, the model's predictions were both confident and reliable. Notably, these results were not just quantitatively superior, they were achieved through a lean architecture that supports scalability and real-time application, fulfilling the project's original goal.

One of the most important insights arose from addressing the class imbalance caused by the overrepresented rest (T0) class. Including T0 during early modeling phases introduced strong class dominance that hindered learning. Removing it not only clarified the decision boundaries between T1 and T2 but also improved the model's capacity to generalize. This strategic adjustment directly answers the fourth research question: excluding the rest class significantly improves classifier focus and accuracy.

While the final model generalized well to unseen validation data, the broader cross-subject analysis revealed variability in individual accuracies. This speaks to the challenge posed by subject heterogeneity, a known limitation in EEG-based classification. On average, subject-specific performance hovered around 51.8%, reinforcing the need for personalization techniques. This partially addresses the fifth research question and suggests that while general models are feasible, they must be supplemented with adaptive mechanisms or fine-tuning strategies to account for inter-subject variability.

In sum, this project provides strong evidence that deep learning, particularly convolutional models, can effectively decode motor intentions from non-invasive EEG signals. It also highlights the importance of thoughtful preprocessing, label design, and model simplicity. These findings not only answer the research questions posed at the outset but also offer a pathway forward for future real-time brain-computer interface systems that rely on interpretable and efficient neural decoding.

10 Future Research Directions

This project uncovered several opportunities for continued investigation and improvement. One of the most pressing challenges observed was the variability in performance across subjects, suggesting the need for more personalized modeling strategies. Future work could explore subject-specific fine-tuning, domain adaptation, or meta-learning techniques that adjust model parameters to individual EEG profiles while retaining the generality learned from broader training data.

Another direction involves improving robustness across sessions and hardware configurations. EEG signals are known to vary not only between individuals but also across different acquisition devices and sessions. Implementing transfer learning, where a model is pretrained on a larger EEG corpus and fine-tuned on a task-specific dataset, may enhance generalization under such conditions.

Additionally, this study relied on a fixed temporal window (from 1 to 4.1 seconds post-cue) for trial segmentation. Future research might examine whether shorter or adaptive windows could reduce latency in real-time systems without sacrificing performance. Similarly, while this work focused on a binary classification task, expanding the model to support multi-class prediction, including rest or other motor imagery classes, would make the system more practical for broader BCI applications. Designing a hierarchical decision structure may help manage increased complexity in such settings.

Finally, real-time deployment represents an essential next step. Testing the model in a live BCI loop would provide insight into how it behaves under naturalistic, dynamic conditions. Coupling real-time inference with feedback mechanisms could foster adaptive learning, making the system more responsive and user-centric. Given the relatively small size of EEG datasets, exploring data augmentation techniques such as time warping, noise injection, or synthetic trial generation via GANs may also improve training stability and reduce overfitting in future iterations.

Together, these directions point toward a robust and deployable EEG classification framework, one that is not only accurate but adaptable, interpretable, and capable of supporting future brain-computer interface technologies.

11 Team Member Role

Angel Barrera
Computer Engineer
Master's Student in Applied Data Science
East Tennessee State University

12 Things I Learned from This Project

This project was a valuable opportunity to deepen my understanding of EEG signal processing, machine learning, and deep learning architectures in the context of brain-computer interface applications. Through hands-on experimentation, I learned how challenging EEG data can be to work with due to its low signal-to-noise ratio, inter-subject variability, and high dimensionality. These challenges pushed me to think critically about preprocessing strategies, data balancing, and model design.

One of the most impactful lessons came from evaluating traditional machine learning approaches. Although methods like CSP and PCA are commonly used in the literature, I observed firsthand how their performance deteriorates when applied to real-world, noisy datasets without sufficient temporal modeling. This motivated my shift toward end-to-end deep learning models, where I gained practical experience designing, training, and validating convolutional neural networks specifically tailored for EEG signals.

I also gained a stronger appreciation for the role of model interpretability, class imbalance, and cross-subject generalization, concepts that extend beyond this project and are applicable to many real-world machine learning tasks. Beyond the technical implementation, I developed better habits around experimentation, documentation, and debugging deep learning pipelines.

Most importantly, this project reaffirmed my interest in neural decoding and non-invasive brain-computer interfaces. It showed me that even relatively lightweight models can yield strong performance if they are well-matched to the structure of the data. I leave this project with a clearer understanding of how to approach EEG modeling and a strong foundation for future work in neuroscience and machine learning.

References

[Fahsi, 2020] Fahsi, R. (2020). Eeg motor imagery classification. `https://github.com/reshalfahsi/eeg-motor-imagery-classification`. GitHub repository.

[Goldberger et al., 2000] Goldberger, A. L., Amaral, L. A. N., Glass, L., Hausdorff, J. M., Ivanov, P. C., Mark, R. G., Mietus, J. E., Moody, G. B., Peng, C.-K., and Stanley, H. E. (2000). Physiobank, physiotoolkit, and physionet: Components of a new research resource for complex physiologic signals. *Circulation*, 101(23):e215–e220.

[Lawhern et al., 2018] Lawhern, V. J., Solon, A. J., Waytowich, N. R., Gordon, S. M., Hung, C.-T., and Lance, B. J. (2018). Eegnet: A compact convolutional neural network for eeg-based brain–computer interfaces. *Journal of Neural Engineering*, 15(5):056013.

[Pfurtscheller and Lopes da Silva, 1999] Pfurtscheller, G. and Lopes da Silva, F. H. (1999). Event-related eeg/meg synchronization and desynchronization: basic principles. *Clinical neurophysiology*, 110(11):1842–1857.

[Ramoser et al., 2000] Ramoser, H., Muller-Gerking, J., and Pfurtscheller, G. (2000). Optimal spatial filtering of single trial eeg during imagined hand movement. *IEEE Transactions on Rehabilitation Engineering*, 8(4):441–446. `https://www.cs.hmc.edu/~keller/eeg/Ramoser.pdf`.

[Schalk et al., 2004] Schalk, G., McFarland, D. J., Hinterberger, T., Birbaumer, N., and Wolpaw, J. R. (2004). Bci2000: A general-purpose brain-computer interface (bci) system. *IEEE Transactions on Biomedical Engineering*, 51(6):1034–1043.

Appendix: Figures and Visualizations

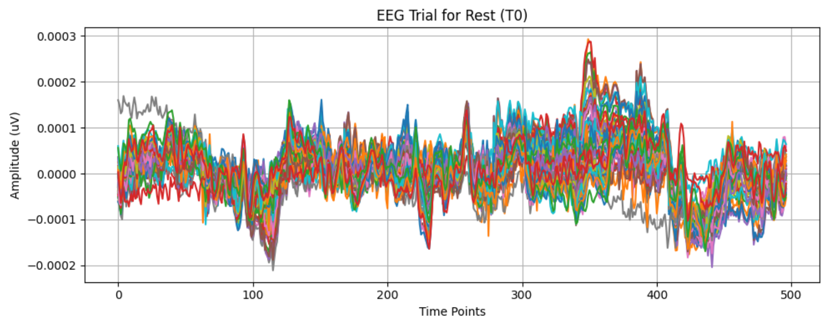

Figure A1: Sample EEG trials labeled as Rest (T0).

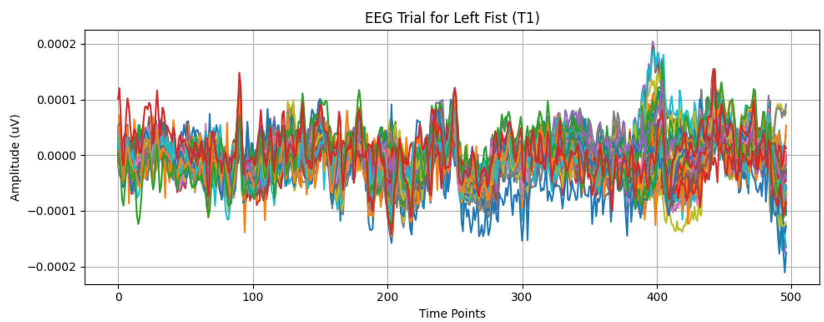

Figure A2: Sample EEG trials for Imagined Left Fist Movement (T1).

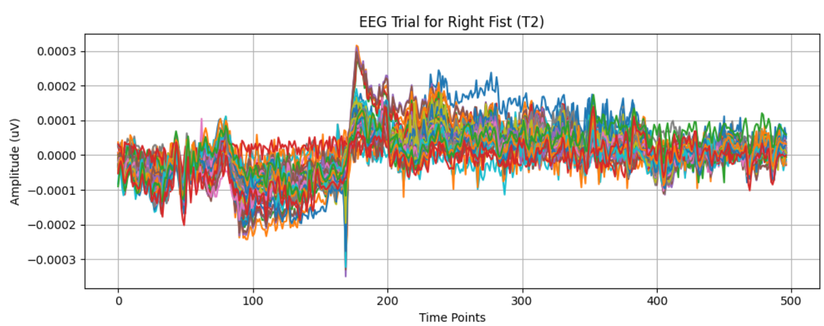

Figure A3: Sample EEG trials for Imagined Right Fist Movement (T2).

```
Classification Report:
              precision    recall  f1-score   support

   Left Fist       0.50      0.46      0.48       106
  Right Fist       0.50      0.54      0.52       106

    accuracy                           0.50       212
   macro avg       0.50      0.50      0.50       212
weighted avg       0.50      0.50      0.50       212

Confusion Matrix:
[[49 57]
 [49 57]]
```

Figure A4: Pipeline visualization for CSP + Logistic Regression.

```
Accuracy for 8 components: 0.5000

--- Testing CSP with 12 components ---
Computing rank from data with rank=None
    Using tolerance 0.0029 (2.2e-16 eps * 64 dim * 2.1e+11  max singular value)
    Estimated rank (data): 64
    data: rank 64 computed from 64 data channels with 0 projectors
Reducing data rank from 64 -> 64
Estimating class=0 covariance using EMPIRICAL
Done.
Estimating class=1 covariance using EMPIRICAL
Done.
Accuracy for 12 components: 0.4481

--- Testing CSP with 16 components ---
Computing rank from data with rank=None
    Using tolerance 0.0029 (2.2e-16 eps * 64 dim * 2.1e+11  max singular value)
    Estimated rank (data): 64
    data: rank 64 computed from 64 data channels with 0 projectors
Reducing data rank from 64 -> 64
Estimating class=0 covariance using EMPIRICAL
Done.
Estimating class=1 covariance using EMPIRICAL
Done.
Accuracy for 16 components: 0.4953
```

Figure A5: Accuracy for different CSP component settings (8, 12, 16).

```
Classification Report:
              precision    recall  f1-score   support

   Left Fist       0.50      0.46      0.48       106
  Right Fist       0.50      0.54      0.52       106

    accuracy                           0.50       212
   macro avg       0.50      0.50      0.50       212
weighted avg       0.50      0.50      0.50       212

Confusion Matrix:
[[49 57]
 [49 57]]
```

Figure A6: CSP component regularization and empirical covariance estimation.

```
--- Random Forest ---
Classification Report:
              precision    recall  f1-score   support

   Left Fist       0.48      0.45      0.47       106
  Right Fist       0.48      0.51      0.50       106

    accuracy                           0.48       212
   macro avg       0.48      0.48      0.48       212
weighted avg       0.48      0.48      0.48       212

Confusion Matrix:
[[48 58]
 [52 54]]
```

Figure A7: Results from Random Forest classifier trained on CSP + PCA features.

```
--- Gradient Boosting ---
Classification Report:
              precision    recall  f1-score   support

   Left Fist       0.48      0.49      0.49       106
  Right Fist       0.48      0.47      0.48       106

    accuracy                           0.48       212
   macro avg       0.48      0.48      0.48       212
weighted avg       0.48      0.48      0.48       212

Confusion Matrix:
[[52 54]
 [56 50]]
```

Figure A8: Results from Gradient Boosting classifier trained on CSP + PCA features.

```
--- SVM (RBF Kernel) ---
Classification Report:
              precision    recall  f1-score   support

   Left Fist       0.48      0.65      0.55       106
  Right Fist       0.45      0.28      0.35       106

    accuracy                           0.47       212
   macro avg       0.46      0.47      0.45       212
weighted avg       0.46      0.47      0.45       212

Confusion Matrix:
[[69 37]
 [76 30]]
```

Figure A9: SVM (RBF kernel).

	precision	recall	f1-score	support
Left Fist	0.83	0.43	0.57	23
Right Fist	0.61	0.91	0.73	22
accuracy			0.67	45
macro avg	0.72	0.67	0.65	45
weighted avg	0.72	0.67	0.65	45

Figure A10: CNN + Transformer hybrid model architecture. The model combines a Conv1D layer with a self-attention mechanism for motor intention decoding.

Classification Report:

	precision	recall	f1-score	support
Left Fist	0.71	0.87	0.78	23
Right Fist	0.82	0.64	0.72	22
accuracy			0.76	45
macro avg	0.77	0.75	0.75	45
weighted avg	0.77	0.76	0.75	45

Figure A11: Architecture of the final CNN model using two stacked Conv1D layers.

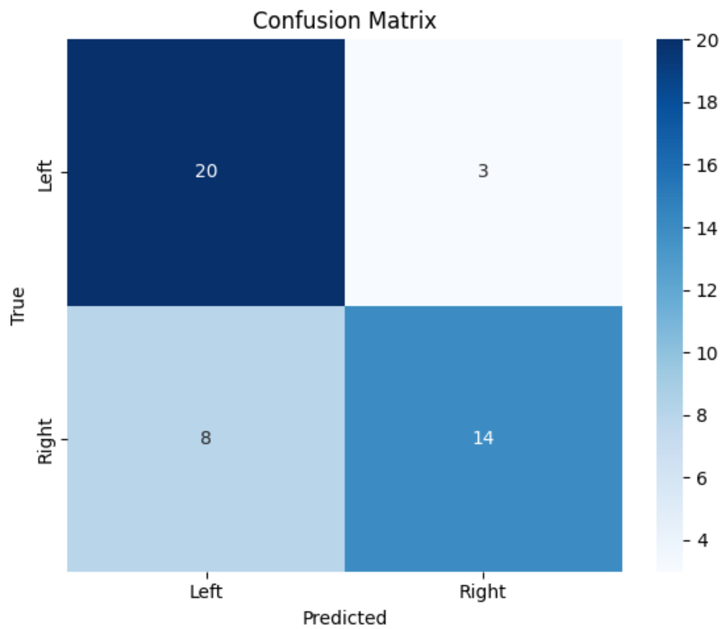

Figure A12: Confusion matrix for the final CNN classifier on validation data.

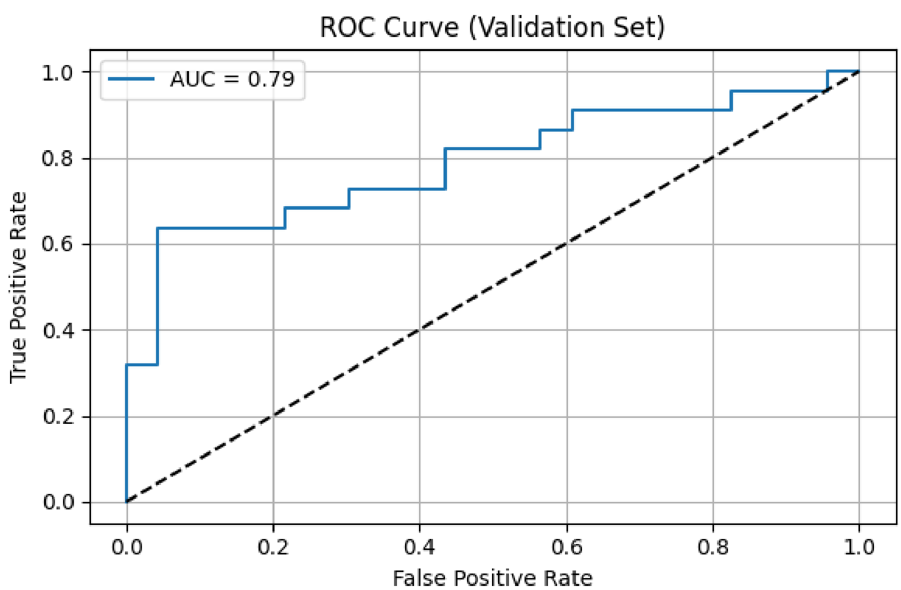

Figure A13: Receiver Operating Characteristic (ROC) curve for final CNN.

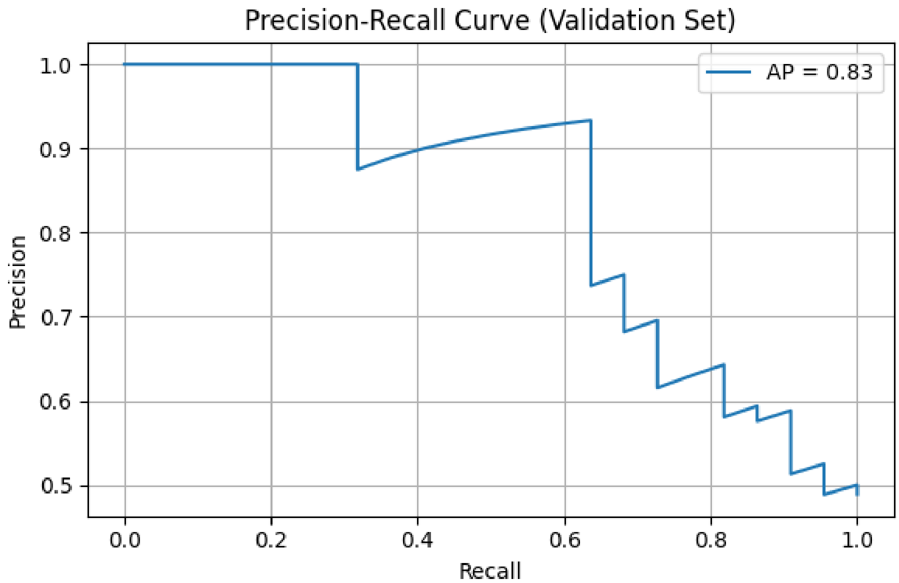

Figure A14: Precision-Recall curve for final CNN.

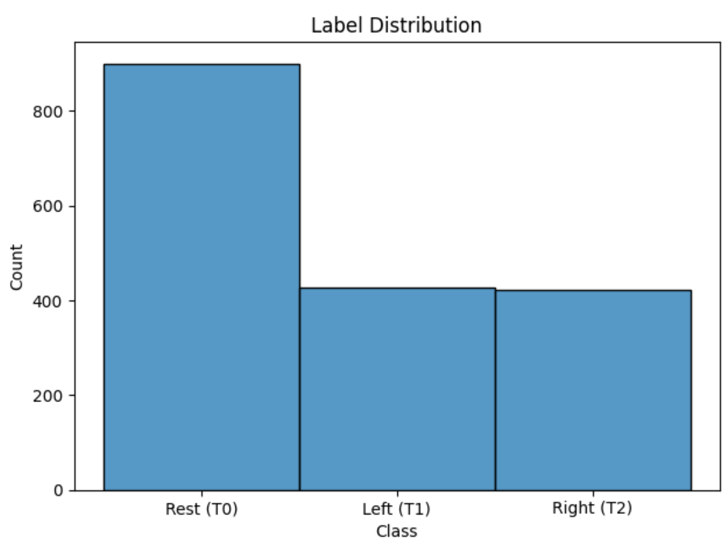

Figure A15: Class distribution in the original dataset showing imbalance, especially in the Rest (T0) class.